BIOT Module 5

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1. Introduction

Two patients receive the same heart medicine after the same procedure. For one, it works as intended. The other stays at higher risk of another heart attack, because their body barely switches the drug on. The difference can come down to a single letter in one gene, CYP2C19.

Genetic tests now report variants like these routinely, but many come back labelled “of uncertain significance”: the lab cannot say whether they matter. Computer tools that predict whether a variant damages a protein are one way to try to settle them.

This module steps away from Alzheimer’s targets to the other end of drug development: how a patient’s own DNA changes the way they respond to a drug.

This brings us to the main question of this module: can variant-effect scores be trusted to classify CYP2C19’s uncertain variants, and what do they miss?

2. Gene and data choice, with biological interpretation

The gene, CYP2C19, was assigned on Canvas. It is a liver enzyme that processes many common drugs: it switches clopidogrel on and breaks down voriconazole, some antidepressants and acid-reflux drugs. It is one of the genes with CPIC dosing guidelines, so getting its variants right changes prescriptions.

The data is ClinVar labels with CADD, REVEL, AlphaMissense and gnomAD frequencies, all fetched through MyVariant.info. This puts the clinical label, three predictors and the population frequency side by side for every variant. All numbers come from my notebook. I skipped the AlphaGenome Atlas because I had no API key.

What the results mean biologically:

- *Known alleles recovered:* the main star alleles are all in the data (*2, *3, *4, *8 and *17), so the fetch captured the variants that clinical tests report.
- *Surprise:* ClinVar has no pathogenic CYP2C19 variants at all. Its broken alleles are filed as “drug response”, and two of them are common (*2 at 16.8%, *17 at 20.0%). A drug gene behaves very differently from a disease gene, which shapes every step below.

Star alleles: versions of a drug gene are named with a star. *1 is the normal version. *2 and *3 make no working enzyme, and *17 makes extra enzyme. Clinical tests and CPIC drug guidelines use these names.

Fetch checks (Q1):

- *Count:* 80 ClinVar variants (64 distinct rsIDs), so the fetch worked.
- *Anchors:* both are present: *2 (rs4244285, frequency 16.8%) and *17 (rs12248560, 20.0%).
- *Scores:* CADD scored 68 of 80 variants, REVEL 54, AlphaMissense 54, and 55 have a gnomAD frequency. No column is empty.
- *Labels:* 37 “drug response”, 33 VUS, 9 benign and 0 pathogenic. That zero matters for everything below.

3. Thresholds and calibration

Frequency threshold: 0.5. CYP2C19 is a drug gene, not a disease gene. Its broken versions don’t make anyone ill; they only matter when a person takes certain drugs. So nothing keeps them rare: *2 is at 16.8% and *17 at 20.0%. The only variants worth removing are ones where the “variant” base is actually the majority base, so I set the cut-off at 0.5.

- *What it removed:* 2 variants, both benign majority bases: I331V (95%) and one silent change (92%).

- *Missing frequencies:* 25 variants have no gnomAD frequency. I kept them as rare, because a plain `gnomad_af <= MAX_AF` test would have dropped them without warning.

The 0.001 contrast: a rare-disease cut-off would remove 19 variants, 14 of them functional, including *2, *17, *3, *4 and *8. It would throw away almost every allele a CYP2C19 test reports.

Predictor: CADD. It is the only tool that scores variants outside the protein code. It covered 85% of the gene, against 67.5% for REVEL and AlphaMissense. That matters here because *2 is a "silent" change and *17 sits in front of the gene.

Calibration on CYP2C19 alone was impossible. It has 0 pathogenic variants, so there is nothing to compare the benign ones against. ClinVar files CYP2C19's broken alleles as "drug response", not "pathogenic". So I took the documented route and pooled with DPYD, a gene with many pathogenic variants:

CADD band	Variants	Pathogenic / benign	LR+	Evidence
0–10	65	0 / 65	0.00	strong, benign
10–20	24	0 / 24	0.00	strong, benign
20–25	11	5 / 6	2.05	none either way
25–30	4	3 / 1	7.38	moderate, pathogenic
30–50	31	31 / 0	inf	strong, pathogenic

How far I trust it: not far.

- *Sample size:* 135 labelled variants (39 pathogenic, 96 benign). The 25–30 band has only 4 variants, so its 7.38 is noise.
- *Empty cells:* LR 0 and infinity just mean one group had no variants in that band, not perfect evidence.
- *Wrong gene:* all 39 pathogenic variants are from DPYD, so the table describes DPYD, not CYP2C19.

I tested it on CYP2C19's own known broken alleles, and it failed:

Allele	Change	CADD	Pooled band says
*2	silent change that breaks splicing	5.7	strong, benign
*4	start codon lost (M1V)	15.9	strong, benign
*6	R132Q	23.0	no evidence
*8	W120R	25.7	moderate, pathogenic
*5	R433W	32.0	strong, pathogenic
*3	early stop (W212X)	35.0	strong, pathogenic
*7	splice site	none	unscored

The pooled bands call *2, the most important broken allele in the gene, "strong, benign". So I can't use them to judge anything else in CYP2C19.

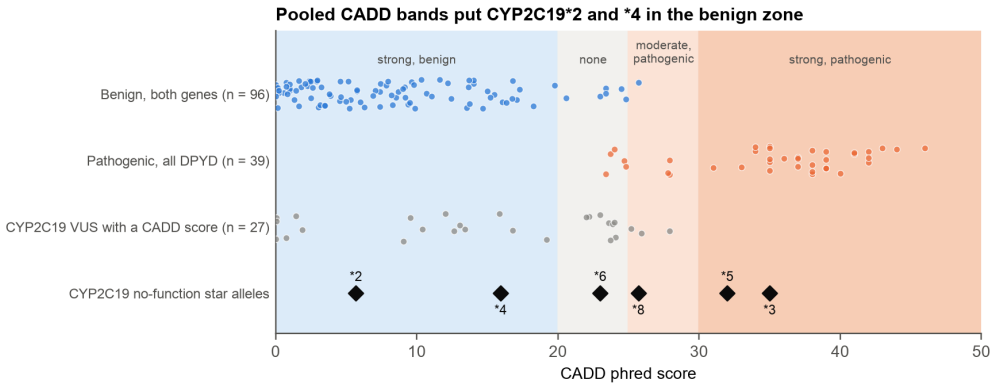


Figure 1: CADD scores of the pooled labelled variants, the CYP2C19 VUS and CYP2C19's own no-function alleles, over the pooled evidence bands. Every pathogenic variant is from DPYD, and *2 and *4 sit in the "strong, benign" band. *7 has no CADD score.

VUS results (Q5):

- *Pooled bands taken at face value:* 16 of the 33 VUS would move towards benign and none towards pathogenic.
- *My call:* none move, and all 33 stay VUS. The 16 "benign" moves all have CADD below 20, the same range where *2 and *4 sit.
- *The other 17:* 8 gave no evidence either way, 3 fell in the 4-variant band, and 6 are large deletions that no tool scored.

4. Coverage gaps and the regulatory genome

What the missense tools could not score: REVEL scored nothing in eight classes, 25 variants in all: large deletions (8), silent changes (7), changes in front of the gene (5), and one each of intergenic, intronic, splice site, early stop and lost stop. CADD scored 14 of the 25. It missed the 8 large deletions (written as ranges, not single bases), the splice-site variant, one silent change and one change in front of the gene.

*One unscored variant: *7 (rs72558186).*

- *What it is:* a change at the edge of an intron that the splicing machinery needs, so it likely breaks splicing.
- *Scores:* no tool scored it. That means not scored, not "benign". Treating the blank as harmless would hide a broken allele.
- *What would classify it:* evidence about splicing. SpliceAI would predict whether the splice site is lost. An RNA test would show it directly: either reading CYP2C19 RNA from a carrier's liver tissue, or testing the variant in cells with a minigene. Both show whether part of the RNA is cut out or kept by mistake.

The Atlas (skipped and disclosed): my gene has 25 non-missense variants, which is exactly what the Atlas is for. It could have told me:

- **17:* whether its effect comes from a liver gene-switch signal (DNase, TF ChIP or RNA-seq). That is how *17 makes extra enzyme.
- **2:* whether the Atlas sees its splicing problem, which CADD missed.
- *The 5 upstream and 1 intronic "drug response" variants:* whether each has its own regulatory signal, or is just inherited alongside *2 or *17.

No other tool in the course reads gene-switch regions in specific tissues.

5. Discordance: CYP2C19*2

Rule it tripped: "common but functional". It is at 16.8%, far above 1%, yet ClinVar treats it as functional.

What each tool said:

- *snpEff:* "silent", because the DNA change (c.681G>A) does not change the amino acid.
- *REVEL and AlphaMissense:* no score, because it is not a protein change.
- *CADD:* 5.7, which looks harmless.
- *ClinVar:* the most important broken allele in the gene.

What would have to be true for all of these to be right: the change must work through RNA, not protein. This is the explanation I argue for. The change creates a new, stronger splice site inside exon 5. The cell uses it and cuts out the first 40 letters of the exon. The rest of the RNA is then read in the wrong frame, so it hits an early stop and no enzyme is made. So:

- snpEff is right that the amino acid is unchanged.
- The missense tools are right that there is no protein change to score.
- CADD simply missed the new splice site.

It stays common because it does no harm until someone takes a drug that needs CYP2C19. I reject the other explanation, that *2 just travels with some other causal change, because the 40-letter cut has been seen directly in liver RNA.

6. Limitations: what I could not determine, and the one experiment

Two things my analysis can't tell me about *2:

1. *Zygosity and phase:* I looked at variants one at a time, but a person has two copies of the gene. One copy of *2 gives an intermediate metabolizer; two copies, or *2 plus *3, give a poor metabolizer. Only testing both copies,

and knowing which variants sit together on each, gives the combination that CPIC guidelines use.

2. *Modifiers (phenoconversion):* another drug can block CYP2C19. Omeprazole, a common acid-reflux drug, does this, so a person with normal genes can behave like a poor metabolizer. DNA alone can't show that.

My most interesting call: keeping 16 VUS as VUS instead of moving them to benign.

- *The experiment that would settle it:* a deep mutational scan, a large lab test that measures the enzyme activity of every possible single amino-acid change in CYP2C19. That would give CYP2C19 its own labels to calibrate CADD on, instead of borrowing DPYD's.
- *What would change my mind:* if nearly all changes with CADD below 20 work normally in that test, I would move the 16 VUS towards benign. If a fair share don't work, keeping them as VUS was right.